# MICHAEL PEARCE

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## ACADEMIC APPOINTMENTS

**Assistant Professor of Statistics (tenure-track)**
Reed College, Department of Mathematics and Statistics 2023–Present

**Visiting Scholar (sabbatical appointment)**
University of California at Berkeley, Department of Statistics 2026–27

## EDUCATION

**University of Washington**, Seattle, WA 2018–23
Ph.D., Statistics (*Statistics in the Social Sciences* track)
Advisor: Elena A. Erosheva
Dissertation: Methods for the Statistical Analysis of Preferences, with Applications to Social Science Data

**St. Olaf College**, Northfield, MN 2013–17
B.A., Mathematics (Statistics concentration)
*Summa cum laude, Phi Beta Kappa*

## PUBLICATIONS

### Refereed Articles

8. **Pearce, M.** and Zhou, S. (2026) "RaCE: a Rank-Clustering Estimation Method for Network Meta-Analysis." *Research Synthesis Methods* 17.2: 314–331.
7. **Pearce, M.** and Erosheva, E.A. (2025) "Bayesian Rank-Clustering" *Psychometrika* 90.3: 904–931.
6. **Pearce, M.** and Perlman, M. (2025) "Estimating the ratio of means in a zero-inflated Poisson mixture model" *Stats* 8.3: 55.
5. **Pearce, M.** and Erosheva, E.A. (2025) "Modeling Preferences: A Bayesian Mixture of Finite Mixtures for Rankings and Ratings" *Journal of the American Statistical Association* 120.551: 1621–1632.
4. Gallo, S.A., **Pearce, M.**, Lee, C.J., and Erosheva, E.A. (2023) "A new approach to peer review assessments: Score, then rank." *Research Integrity and Peer Review* 8.10.
3. **Pearce, M.** and Erosheva, E.A. (2022) "A unified statistical learning model for rankings and scores with application to grant panel review." *Journal of Machine Learning Research* 23.210: 1–33.
2. **Pearce, M.** and Raftery, A.E. (2021) "Probabilistic forecasting of maximum human lifespan by 2100 using Bayesian population projections." *Demographic Research* 44.52: 1271–1294.
1. **Pearce, M.***, Sparrow, Z.*, Mabote, T. R., and Sanchez-Gonzalez, R. (2020) "stoBEST: An efficient methodology for increased spatial resolution in two-component molecular tagging velocimetry." *Measurement Science and Technology* 32.3: 035302. *(*indicates authors contributed equally.)*

### Preprints and Other Scholarly Works

6. **Pearce, M.** (2026) "Efficient Bayesian inference for Benter models on ranked data." *arXiv preprint arXiv:2608.23825.*
5. **Pearce, M.** Lipman, E.R., Adolph, C., and Erosheva, E.A. (2026) "Can statistical models capture Mamdani's success? Social choice, ranked-choice voting, and model fit, with an application to the 2025 New York City Democratic Primary." *arXiv preprint arXiv:2608.23800.*
4. **Pearce, M.** and Zhou, S. (2026) "RaCE.NMA: An R Package for Rank-Clustering Interventions in Network Meta-Analysis." *Preprint available upon request.*
3. **Pearce, M.** and Erosheva, E.A. (2022) "On the validity of bootstrap uncertainty estimates in the Mallows-Binomial model." *arXiv preprint arXiv:2206.12365.*

2. **Pearce, M.** and Raftery, A.E. (2021) "Will this be a record-breaking century for human longevity?" *Significance*.

1. **Pearce, M.** and Raftery, A.E. (2021) "The maximum human life span will likely increase this century, but not by more than a decade" *The Conversation*.

## SOFTWARE

**RaCE.NMA: Rank-Clustering Estimation of treatments in a Network Meta-Analysis.** R package available on CRAN. Vignettes available here.

**rankclust: Fit a Bayesian, Rank-Clustered Bradley-Terry-Luce Model to Ordinal Comparison Data.** R package available on Github. Vignettes available here.

**rankrate: Statistical Tools for Preference Learning with Rankings and Ratings.** R package available on CRAN. Vignettes available here.

**Peer Review with Rankings and Ratings.** R Shiny application.

## RESEARCH PRESENTATIONS

27. "What do voters really want? Statistics meets ranked-choice voting" *Macalester College, Department of Mathematics, Statistics, and Computer Science*, St. Paul, MN, October 2026. (*invited colloquium*)

26. "What do voters really want? Statistics meets ranked-choice voting" *St. Olaf College, Department of Mathematics, Statistics, and Computer Science*, Northfield, MN, October 2026. (*invited colloquium*)

25. "Bayesian rank-clustering, with applications to ranked-choice voting and survey data" *San José State University, Mathematics and Statistics Department*, San Jose, CA, September 2026. (*invited colloquium*)

24. "Bayesian rank-clustering of treatments from analyses of clinical trials data" *Working Group on Model-Based Clustering*, Edmonton, AB, July 2026. (*invited session*)

23. "Bayesian rank-clustering, with applications to survey and election data" *ISBA World Meeting*, Nagoya, Japan, July 2026. (*contributed session*)

22. "Ranked-choice voting and mixture modeling: An application to the 2025 New York City Democratic primary" *Classification Society Annual Meeting*, Halifax, NS, June 2026. (*invited session*)

21. "Rank-clustering treatments in network meta-analyses of clinical trials data" *Oregon Chapter of the ASA Fall Meeting*, Portland, OR, November 2026. (*invited speaker*)

20. "Bayesian Rank-Clustering" *15th Scientific Meeting Classification and Data Analysis Group*, Naples, Italy, September 2025 (*invited session*)

19. "Rank-clustering treatments in network meta-analyses of clinical trials data" *Classification Society Annual Meeting*, Ottawa, ON, June 2025. (*contributed session*)

18. "Statistical Perspectives on Ranked-Choice Voting" *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, February 2025. (*invited speaker*)

17. "Statistical Estimation with Ranked Choice Voting" *University of Washington*, Seattle, WA, October 2024. (*invited lecture in STAT 452*)

16. "Bayesian Rank-Clustering" *Washington State University Vancouver*, Vancouver, WA, October 2024. (*invited seminar*)

15. "Bayesian Rank-Clustering" *Penn State University; Health Science Department*, University Park, PA, October 2024. (*invited seminar*)

14. "Bayesian Rank-Clustering" *Reed College*, Portland, OR, September 2024. (*colloquium*)

13. "Broadening Access to Bayesian Statistics with Active Learning" *Joint Statistical Meetings*, Portland, OR, August 2024. (*contributed session*)

12. "Bayesian Rank Clustering" *ISBA World Meeting,* Venice, Italy, July 2024. (*poster*)

11. "Methods for the Statistical Analysis of Preferences, with Applications to Social Science Data" *Classification Society Annual Meeting*, Kelowna, BC, June 2024. (*Distinguished Dissertation Award* session).

10. "Bayesian Rank Clustering" *Center for Statistics in the Social Sciences 25th Anniversary Conference* Seattle, WA, May 2024. (*poster*)
9. "Weighted Mallows-Binomial Model for Rankings and Ratings" *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, April 2024. (*invited speaker*)
8. "Rankings and ratings in peer review: A mixture of finite mixtures accounting for degree of reviewer leniency" *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, November 2023. (*invited speaker*)
7. "Improving preference analysis: Joint models for ordinal and cardinal data" *International Meeting of the Psychometrics Society*, College Park, MD, July 2023. (*contributed session*)
6. "A Unified Statistical Learning Model for Rankings and Scores with Application to Grant Panel Review" *NeurIPS*, New Orleans, LA, December 2022. (*poster*)
5. "Fast Bayesian estimation for ranking models" *Joint Statistical Meetings*, Washington, DC, August 2022. (*contributed session*)
4. "Using ranking data for decision-making" *Joint Statistical Meetings*, Washington, DC, August 2022. (*contributed session chair*)
3. "Joint Bayesian inference for rankings and ratings under heterogeneous preferences" *ISBA World Meeting,* Montreal, QC, June 2022. (*poster*)
2. "Unified latent class modeling of scores and rankings applied to grant panel review" *Working Group on Model-Based Clustering*, Athens, Greece, October 2021. (*contributed session*)
1. "Unified latent class modeling of score and rank data applied to grant panel review" *Joint Statistical Meetings*, Seattle, WA, August 2021. (*contributed session*)

## SELECTED MEDIA COVERAGE

- *Washington Post* "Want to add healthy years to your life? Here's what new longevity research says." (Oct. 11, 2021)
- *BBC News (Brazil)* "Por que cada vez mais pessoas estão vivendo até os 100 anos?" (July 11, 2022)
- *CNBC* "Researchers say the probability of living past 110 is on the rise — here's what you can do to get there" (July 17, 2021)
- *Stats and Stories (Podcast)* "The Age of the Supercentenarian" (April 29, 2022)
- *Southern Weekly (China)* "What is the limit of human life span?" (Sept. 16, 2021)
- *Elemental (Medium)* "How Long Can Humans Really Live?" (July 15, 2021)
- *Gulf News* "Surviving up to 150: How long can a person live?" (July 12, 2021)
- *Indian Express* "Can a person live to age 124, 135 or 150? Some optimism, some caveats" (July 6, 2021)
- *The South African* "Rise of the supercentenarians: Today's kids could live for 130 years" (July 4, 2021)
- *UW News* "How long can a person live? The 21st century may see a record-breaker" (July 1, 2021)

## UNDERGRADUATE THESIS ADVISING

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|---|---|
| Conor Bekaert, *Towards a Weighted Joint Statistical Model for Rankings and Ratings* | 2024 |
| Quinn Hargrove, *Visualizations to Improve Ranked Data Analyses, with Applications to Board Game Data* | 2024 |
| Chance Addis, *Kingmaker: A Simulation Analysis of Strategic Voting* | 2025 |
| Emma Dunn, *A Bayesian Methodology for Forecasting Instant-Runoff Elections* | 2025 |
| Tory Norton, *How Many Mixtures? Frequentist and Bayesian Approaches to Inferring the Number of Latent Groups* | 2025 |
| Chris Li, *Evaluating Large Language Models for Data Science Tasks in R* | 2026 |
| Nathan Holland, *Spatial Statistics for Environmental Policy, with Application to Air Quality in Los Angeles* | 2026 |
| Deltas Shi, *Inferring Team Strength in Ultimate Frisbee: A Comparative Study of Ranking Systems* | 2026 |

## TEACHING

**Reed College**

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|---|---|
| Instructor, STAT 141 (Introduction to Probability and Statistics) | Fall 2023, Fall 2024 |
| Instructor, STAT 243 (Statistical Learning) | Spring 2025, Fall 2025 |
| Instructor, STAT 343 (Statistics Practicum) | Fall 2025 |
| Instructor, STAT 346 (Bayesian Statistics) | Spring 2024, Spring 2026 |
| Instructor, STAT 392 (Mathematical Statistics) | Spring 2025, Spring 2026 |

**University of Washington**

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|---|---|
| Instructor, CSSS 508 (Introduction to R for Social Scientists) | Autumn 2022, Spring 2023 |
| Teaching Assistant, CSSS 589 (Multivariate Data Analysis for the Social Sciences) | Autumn 2021 |
| Teaching Assistant, CSSS 594 (Statistics and Philosophy of Voting) | Autumn 2020 |
| Teaching Assistant, STAT 220 (Statistical Reasoning) | Autumn 2019 |
| Teaching Assistant, STAT 311 (Elements of Statistical Methods) | Autumn 2018, Winter 2019 |
| Teaching Assistant, STAT 342 (Introduction to Probability and Mathematical Statistics) | Spring 2019 |
| Teaching Assistant, STAT 528 (Applied Statistics Capstone) | Winter 2021, Winter 2022 |
| Mentor, Directed Reading Program | 7 quarters; Autumn 2020–Spring 2022 |
| Mentor, Washington eXperimental Mathematics Lab | Autumn 2021 |

## SERVICE

**Reed College**

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| Statistics Program Chair | 2025–26 |
| Library Board | 2024–25, 2025–26 |
| Undergraduate Research Committee | 2023–24 |
| Math and Statistics Department Colloqium organizer | 2023–24 |

**University of Washington**

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| Pre-application review service reviewer | 2023 |
| Queer Union for (Bio)statistician Inclusion and Community affinity group founder | 2022–23 |
| Undergraduate statistics curriculum committee | 2021–22 |
| PhD program admissions reviewer | 2020–21 |

**Broader Scholarly Community**

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| Bayesian Education Research and Practice (BERaP) ISBA section treasurer | 2026–27 |
| Scientific organizing committee for the Classification Society Annual Meeting | 2026 |
| Undergraduate Statistics Project Competition judge | 2023, 2025 |
| International Conference on Machine Learning workflow chair | 2021 |
| Manuscript review (various journals, e.g., *Annals of Applied Statistics, Journal of Survey Statistics and Methodology, Journal of Statistics and Data Science Education, Journal of Open Research Software, Biomedicine Hub, Cogent Social Sciences*) | |

## HONORS AND AWARDS

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| **Mitchell Prize** *(International Society for Bayesian Analysis)* | 2025 |
| **Distinguished Dissertation Award** *(The Classification Society)* | 2024 |
| **Best Poster Award** *(International Society for Bayesian Analysis World Meeting)* | 2024 |
| **Travel Award** *(International Society for Bayesian Analysis)* | 2022, 2024 |
| **Scholar Award** *(NeurIPS)* | 2022 |
| **Dorothy M. Gilford Teaching Award** *(University of Washington)* | 2021 |

## INDUSTRY EXPERIENCE

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|---|---|
| **Boeing Research and Technology** (*Applied Statistics PhD Intern*) | 2019–2020 |
| **Deloitte LLC** (*Analytics Consultant*) | 2017–2018 |

## PROFESSIONAL AFFILIATIONS

American Statistical Association
International Society for Bayesian Analysis
The Classification Society